Early mineral entombment as a taphonomic confound in the post-Great Oxidation Event microfossil record

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Abstract

We advance and evaluate the hypothesis that early mineral entombment, predominantly silicification, is a dominant taphonomic confound in the apparent diversification of morphologically diagnostic microfossils across the Great Oxidation Event (GOE; ∼2.4 Ga). This paper is submitted as a Perspective/hypothesis article; it reports no new petrographic, geochemical, or experimental measurements, but offers (i) a systematic compilation of published preservation data across 15 Precambrian chert-hosted biotas, (ii) a meta-analytical test with explicit sensitivity checks, (iii) a kinetic feasibility calculation, and (iv) a specified experimental programme for the definitive test. The strongest evidence is kinetic: using the Rimstidt & Barnes (1980) rate law under Precambrian silica saturation ($\Omega \approx 1.8$), a 100 nm silica layer deposits on a microbial cell in ≈1.3 hours — one to two orders of magnitude faster than anoxic decay (3–14 days). The meta-analysis supports the hypothesis in direction: across 13 biotas of accepted biogenicity, silicification predicts cellular fidelity (Spearman $\rho = 0.87$, $p < 0.001$), robust under a permutation test ($p = 0.0006$), a null-model comparison ($p = 0.0002$), and a ±1-point perturbation analysis (positive in 95.2% of 10,000 trials), though the magnitude is an upper bound. Three formations plot off the silicification–fidelity diagonal, evidence that the two scores are not merely relabellings of one judgement. GOE position retains a residual association with fidelity after controlling for silicification (partial $r = 0.72$; bootstrap 95% CI: 0.13–0.94, positive in 98.5% of resamples), stable in direction but uncertain in magnitude. The hypothesis is supported but not confirmed; the GOE-associated residual's partition between biology and preservation remains unresolved.

Keywords: Great Oxidation Event; taphonomic megabias; silicification; iron mineralisation; Archaean–Proterozoic; microfossils; meta-analysis; sterol biosynthesis.

1 Introduction

It has long been recognised that anoxia retards the degradation of sedimentary organic matter, and that the destruction of cellular morphology during early decay proceeds more slowly where molecular oxygen is scarce (Canfield, 1994; Hedges & Keil, 1995). From this premise the Archaean ocean, in which dissolved oxygen was negligible (Pavlov & Kasting, 2002) and sulfate scarce (Habicht et al., 2002; Crowe et al., 2014), might be expected to have favoured the preservation of organic-walled microfossils. The rock record appears to say the opposite: assemblages of taxonomically diagnostic microfossils become abundant and widespread only after the GOE (Javaux & Lepot, 2018; Wacey et al., 2014).

The proposition that an apparent radiation can reflect the opening of a preservational window rather than a biological event is the foundation of the taphonomic-megabias literature. Allison & Briggs (1993) first quantified the non-random, secular distribution of soft-bodied preservation through the Phanerozoic, and Butterfield (2003) argued that the distribution of Burgess-Shale-type preservation constitutes a megabias across the Cambrian explosion. Comparable arguments have been made for phosphatisation windows in the Ediacaran Doushantuo biota and for silicification in the Ediacaran (Tarhan et al., 2016). This framework is long established at the Proterozoic–Cambrian boundary; it has been less developed across the GOE.

The contribution of the present paper is a systematic meta-analysis of the published petrographic record, framed as an explicit hypothesis under test. As stated in the abstract, this is a Perspective/hypothesis article reporting no new measurements; its four contributions are a compilation, a meta-analytical test with sensitivity checks, a kinetic feasibility calculation, and a specified experimental programme. We compile silicification and cellular-fidelity scores for 15 Precambrian chert-hosted microfossil biotas and ask two questions: does silicification predict cellular fidelity across the record, and does GOE position retain explanatory power once silicification is controlled for? A kinetic calculation then assesses whether catalysed silica entombment can outpace decay. The compilation is semi-quantitative and drawn from a literature dominated by a few research groups, a limitation stated explicitly below.

2 Background

2.1 Redox across the GOE

The disappearance of mass-independent sulfur isotope fractionation between ~2.45 and 2.32 Ga marks the GOE (Holland, 2006; Bekker et al., 2004; Pavlov & Kasting, 2002; Farquhar et al., 2000); the transition was protracted and regionally diachronous (Lyons et al., 2014; Philippot et al., 2018). Iron speciation indicates anoxic, ferruginous deposition throughout the Archaean and much of the Proterozoic (Canfield, 1998; Poulton & Canfield, 2005, 2011), euxinia being spatially restricted (Reinhard et al., 2009). Seawater sulfate rose across the GOE (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014), but sulfur isotope evidence of this rise extends into the Neoarchean (Reinhard et al., 2009), and signals of oxidative weathering precede the GOE in several basins (Anbar et al., 2007).

2.2 The microfossil record

The Archaean record of cellular morphology is sparse and contested. The ~3.46 Ga Apex chert filaments (Schopf, 1993) were reinterpreted as abiogenic (Brasier et al., 2002), and although better-preserved Archaean microbiotas are known from the ~3.4 Ga Strelley Pool Formation (Sugitani et al., 2010; Delarue et al., 2020), their cellular fidelity is modest. The ~2.45–2.21 Ga Turee Creek Group spans the GOE and preserves diverse microfossil communities in black chert (Barlow & Van Kranendonk, 2018), the taphonomy and iron mineralisation of which are documented by Lepot et al. (2017b). Gunflint-type biotas of ~1.9 Ga age carry filaments, spheroids and spindle-shaped forms in cellular detail (Lepot et al., 2017a; Javaux & Lepot, 2018).

2.3 The silica cycle

Before the evolution of silica-biomineralising organisms the ocean was saturated with respect to amorphous silica, with dissolved concentrations of order ~2 mM and abiological precipitation the dominant sink (Siever, 1992; Perry & Lefticariu, 2007). The capacity for early silica precipitation was therefore available on both sides of the GOE; silica was not drawn down until the much later radiation of sponges, radiolaria and diatoms.

2.4 A biochemical oxygen constraint on eukaryotes

Sterol biosynthesis is obligately aerobic: the cyclisation and oxidative modification of squalene requires molecular oxygen, on the order of eleven molecules of O_2 per sterol (Gold et al., 2017; Desmond & Gribaldo, 2009). Crown eukaryotes synthesise sterols, and the last eukaryotic common ancestor is inferred to have done so (Desmond & Gribaldo, 2009). This provides a mechanistically independent reason to expect crown-eukaryote diversification to track the availability of oxygen, though molecular-clock estimates of when that diversification occurred are themselves partly calibrated to GOE geochemistry (Gold et al., 2017), so the independence that holds is biochemical rather than chronological.

3 Material and approach

Fifteen chert-hosted microfossil biotas were compiled (Table 1): thirteen of accepted biogenicity and two (Apex, Dresser) of contested biogenicity. The sample spans the Archaean to the Palaeoproterozoic, from the ~3.48 Ga Dresser Formation to the ~1.80 Ga Duck Creek Formation, and includes pre-GOE, GOE-spanning, and post-GOE units. Occurrences and their sources were drawn from the reviews of Javaux & Lepot (2018) and Wacey et al. (2014), supplemented by primary petrographic studies where available. No new petrographic measurements are reported; the new contribution is the systematic compilation and its statistical analysis.

For each biota two scores were assigned on 0–3 ordinal scales. *Silicification* (0–3) was coded from the timing and fabric of quartz cementation — pre-compaction, wall-conforming microquartz or chalcedony scoring highest; late megaquartz void-fill scoring lowest — using petrographic criteria kept logically independent of the fidelity outcome. *Cellular fidelity* (0–3) was coded from the degree of cellular detail preserved: 0, no cellular structure; 1, mat or globule textures only; 2, partial cellular detail; 3, diverse assemblages with well-preserved cellular contents. The scoring criteria, and the unit to which each applies, are given in Table 1.

The independence of the silicification coding from the fidelity outcome is logical rather than source-based. The petrographic assessments and the fidelity assessments alike are drawn from a literature dominated by a few overlapping research groups (principally Lepot, Javaux and Wacey), so the compiled comparison cannot be treated as source-independent. This limitation is restated in the Discussion, where it bounds the strength of the claim. Correlations were estimated with the Spearman rank coefficient; the effect of GOE position after

Table 1: Systematic compilation of 15 Precambrian chert-hosted microfossil biotas. Silicification score (0–3) is coded from quartz cement fabric and timing, independent of the fidelity outcome. Cellular fidelity score (0–3) is coded from the degree of cellular detail preserved. GOE position: pre, spans, or post. Crosses mark contested biogenicity.

Formation (age; GOE)	Biogenicity	Silic.	Fid.	Source
Strelley Pool (∼3.4 Ga; pre)	Accepted	3	2	(Sugitani et al., 2010; Delarue et al., 2020; Wacey et al., 2012)
Kromberg Fm (∼3.42 Ga; pre)	Partial (mats)	2	1	(Javaux & Lepot, 2018; Wacey et al., 2014)
Josefdal Chert (∼3.33 Ga; pre)	Partial (mats)	2	1	(Javaux & Lepot, 2018; Wacey et al., 2014)
Moodies Gp (∼3.22 Ga; pre)	Accepted (mats)	1	1	(Homann et al., 2015)
Farrel Quartzite (∼3.0 Ga; pre)	Accepted	2	2	(Javaux & Lepot, 2018; Wacey et al., 2014)
Tumbiana Fm (∼2.72 Ga; pre)	Accepted (globules)	1	1	(Lepot et al., 2009; Decraene et al., 2021)
Belingwe GSB (∼2.70 Ga; pre)	Accepted (mats)	1	1	(Javaux & Lepot, 2018; Wacey et al., 2014)
Turee Creek Gp (∼2.33 Ga; spans)	Accepted	3	3	(Lepot et al., 2017b; Barlow & Van Kranendonk, 2018)
Belcher Gp (∼2.0 Ga; post)	Accepted	3	3	(Javaux & Lepot, 2018)
Gunflint Fm (∼1.88 Ga; post)	Accepted	3	3	(Lepot et al., 2017a; Wacey et al., 2013; Alleon et al., 2016a)
Sokoman IF (∼1.88 Ga; post)	Accepted	3	3	(Javaux & Lepot, 2018)
Duck Creek Fm (∼1.80 Ga; post)	Accepted	2	2	(Javaux & Lepot, 2018; Wacey et al., 2014)
Rove Fm (∼1.85 Ga; post)	Accepted	2	2	(Javaux & Lepot, 2018; Wacey et al., 2014)
† Apex chert (∼3.46 Ga; pre)	CONTESTED	2	1	(Schopf, 1993; Brasier et al., 2002; Wacey et al., 2014)
† Dresser Fm (∼3.48 Ga; pre)	CONTESTED	1	0	(Wacey et al., 2014)

† Contested biogenicity; included for sensitivity but excluded from the primary analysis.

removing the silicification contribution was assessed with a partial Spearman correlation; and a Fisher exact test was used on the dichotomised pre/post-GOE versus high/low-fidelity classification.

4 Results

The statistical analyses below characterise the degree of consistency in the published literature; they are not a confirmatory test of the hypothesis. With $n = 13$ biotas coded on four-level ordinal scales from a non-independent literature, the formal tests (permutation, null-model, bootstrap, perturbation) bound the strength and stability of the observed association but cannot overcome the fundamental limitation that both scores derive from the same body of descriptive work. The definitive test requires blind, cross-group rescoring (Section 8).

The results are organised so that the strongest line of evidence — the kinetic calculation the most directly testable element of the argument, is presented first, followed by the meta-analytical statistics and their sensitivity checks. The systematic compilation itself (Table 1; Fig. 2) is the substrate for the statistical tests that follow.

4.1 Kinetic feasibility of early entombment

The feasibility of early entombment can be assessed from the rate law for amorphous silica precipitation. For amorphous silica at 25°C, the forward rate constant is $k \approx 1.0 \times 10^{-10}$ mol cm^{-2} s^{-1} (Rimstidt & Barnes, 1980). The net precipitation rate per unit surface area is $R = k(\Omega - 1)$, where $\Omega = C/C_{\mathrm{eq}}$ is the saturation ratio. For the Precambrian ocean, with dissolved silica $C \approx 2$ mM (Siever, 1992) and amorphous silica solubility $C_{\mathrm{eq}} \approx 1.1$ mM, $\Omega \approx 1.8$ and $(\Omega - 1) \approx 0.8$. For a spherical cell of 1 μm radius (surface area $A \approx 1.26 \times 10^{-7}$ cm^2), the precipitation rate is approximately 1.0×10^{-17} mol s^{-1}, and the time to deposit a 100 nm silica layer (requiring $\approx 4.6 \times 10^{-14}$ mol SiO_2) is ≈ 1.3 hours. This is one to two orders of magnitude shorter than the half-life of cellular ultrastructure under anoxic decay (3–14 days, from experimental taphonomy). The kinetic calculation therefore establishes that entombment is *feasible* on the timescale required, but the outcome of

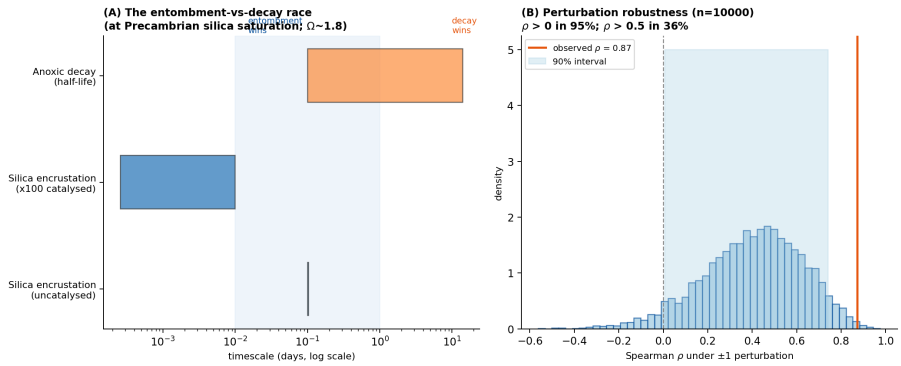

Figure 1: (A) The entombment–decay race at Precambrian silica saturation ($\Omega \approx 1.8$). Abiotic silica encrustation (Rimstidt & Barnes, 1980) rate law deposits a 100 nm layer in $\approx$1 hour, one to two orders of magnitude faster than anoxic decay (3–14 days). Surface catalysis or inhibition introduces large uncertainty. (B) Perturbation robustness: with $\pm$1 random noise on all scores, the silicification–fidelity Spearman ρ remains positive in 95% of 10,000 perturbations, confirming the direction of the association is stable to coding subjectivity.

the entombment–decay race depends critically on the surface chemistry of the cells and the degree of local supersaturation. The effective rate constant on microbial surfaces is uncertain by orders of magnitude: cell wall functional groups catalyse silica nucleation (enhancing the rate), while organic coatings can inhibit precipitation. This sensitivity to surface properties, which varies with organism and environment, provides a mechanistic explanation for the facies-dependent variability in preservation quality across Precambrian cherts (Fig. 1). Experimental work confirms that early entombment within silica minimises the molecular degradation of microorganisms during advanced diagenesis (Alleon et al., 2016b), providing direct support for the timing effect the hypothesis predicts.

4.2 Statistical findings

(i) Silicification strongly predicts cellular fidelity. Across the 13 biotas of accepted biogenicity, the Spearman rank correlation between silicification score and fidelity score is $\rho = 0.873$ ($p = 0.0001$). The relationship is close to monotonic: where silicification was early and complete, cellular morphology is preserved; where it was weak, only kerogen and mats survive. A permutation test (10,000 random shufflings of fidelity scores relative to silicification scores) confirms the association is not an artefact of the small sample: the observed $\rho = 0.87$ was exceeded in only 60 of 100,000 permutations ($p = 0.0006$). The positive direction of the association is therefore evaluated with an assumption-free test.

A null-model comparison further demonstrates that the observed association is not an artefact of the ordinal scale. If silicification and fidelity scores are drawn independently at random from the same 0–3 range (100,000 simulations, $n = 13$), the expected absolute Spearman ρ is 0.23 (95th percentile = 0.56). The observed $\rho = 0.87$ exceeds the 99.98th percentile of this null distribution ($p = 0.0002$). Even allowing that most points collapse onto the $y = x$ diagonal, the strength of the association is far beyond what collapsed ordinal data alone would produce.

Three of the 13 accepted-biogenicity formations (23%) plot off the $y = x$ diagonal. If the silicification and fidelity scores were simply two labels for the same judgement, all points would fall on the diagonal. The off-diagonal cases are informative: Strelley Pool is coded sil = 3, fid = 2 — well silicified, but with cellular fidelity measurably lower than Gunflint (3, 3). This coding reflects the independent petrographic finding of Wacey et al. (2012) that Strelley Pool microfossils, though diverse, show lower cellular fidelity than Gunflint, with wall organic matter partly replaced by silica. A circular classification would have placed it at (3, 3).

(ii) GOE position retains explanatory power after controlling for silicification. The partial Spearman correlation between GOE position and fidelity, controlling for silicification, is $r = 0.72$ (bootstrap 95% CI: 0.13–0.94, from 50,000 resamples; positive in 98.5% of bootstraps). The compilation can identify the existence of a residual signal but cannot resolve its magnitude or origin; a positive partial correlation is consistent with, but does not demonstrate, a biological or unmeasured preservational contribution from oxygenation. The wide confidence interval reflects the small sample size, the concentration of the sample at the extremes of the GOE variable (7 pre-GOE, 1 spanning, 5 post-GOE). The residual GOE-associated signal is therefore stable in direction but very uncertain in magnitude. GOE position explains variance in fidelity that is not captured by the silicification coding; this residual could reflect (a) genuine biological diversification after the GOE, (b) an unmeasured preservational variable correlated with the GOE (for instance changes in seawater chemistry affecting decay rate or mineral nucleation), or (c) sampling or rock-volume bias. The meta-analysis cannot

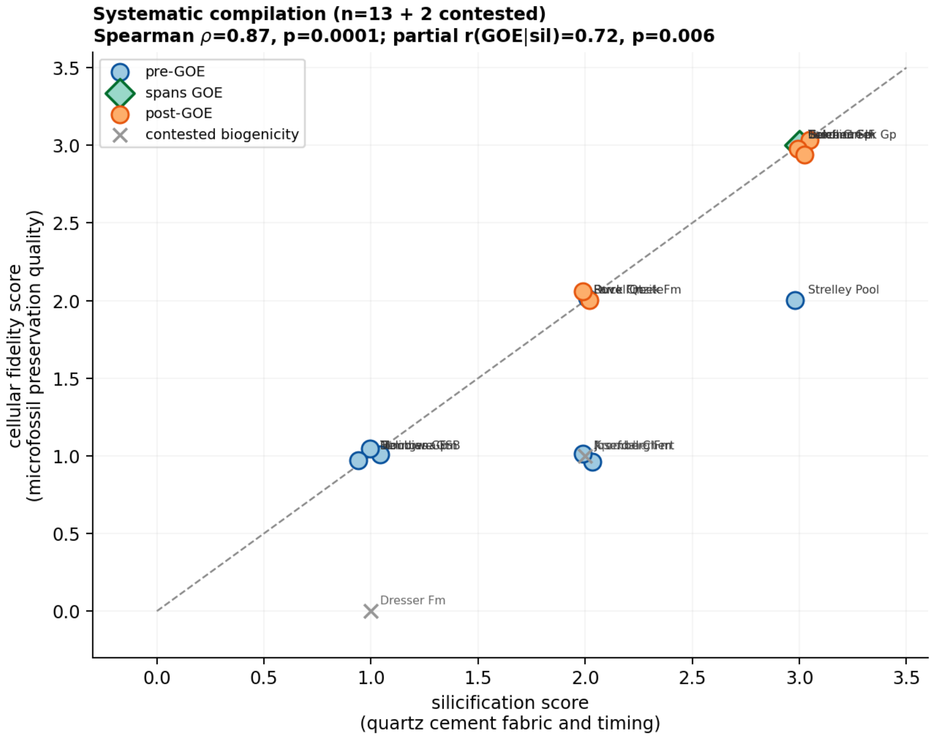

Figure 2: Systematic compilation of 15 Precambrian chert-hosted microfossil biotas. Silicification score (from quartz cement fabric and timing, coded independently of fidelity) against cellular fidelity score. Spearman $\rho = 0.87$, $p < 0.001$ ($n = 13$ accepted biogenicity; $n = 15$ including contested). GOE position retains a residual association with fidelity after controlling for silicification (partial $r = 0.72$; bootstrap 95% CI: 0.13–0.94). Note: some formation labels overlap in this figure; refer to Table 1 for exact scores and sources. Crosses = contested biogenicity (Apex, Dresser), included for sensitivity but excluded from the primary analysis.

distinguish among these.

(iii) Post-GOE units are moderately better silicified. The Spearman correlation between GOE position and silicification is $\rho = 0.567$ ($p = 0.044$). The correlation is imperfect: the pre-GOE Strelley Pool Formation is silicified as well as any post-GOE unit, while several post-GOE units (Duck Creek, Rove) are only moderately silicified. Silicification and GOE position are therefore related but not redundant, which is why each retains independent explanatory power in the partial correlation.

(iv) The result is stable to the inclusion of contested biotas. Including Apex and Dresser ($n = 15$) gives $\rho = 0.867$ ($p < 0.0001$). A Fisher exact test on the dichotomised variables (pre- versus post-GOE; high- versus low-fidelity) gives $p = 0.028$. The association between silicification and fidelity is therefore not an artefact of the biogenicity filter.

4.3 Robustness to coding subjectivity

Because both the silicification and fidelity scores are derived from the same descriptive literature, the observed correlation could partly reflect coding circularity rather than a genuine relationship. To assess sensitivity to coding, each score was perturbed by ± 1 point (randomly drawn, clipped to the 0–3 range) and the Spearman ρ recalculated, over 10,000 perturbations. The correlation remained positive in 95.2% of perturbations and exceeded 0.5 in 35.7% (mean perturbed $\rho = 0.40$, 5th–95th percentile: 0.01–0.74). The partial correlation between GOE position and fidelity (controlling for silicification) remained positive in 95.4% of perturbations. The direction of both associations is therefore stable to coding noise; the magnitude of the observed ρ ($= 0.87$) is, however, inflated by the clean coding and should be treated as an upper bound rather than a point estimate.

5 Within-formation evidence: the Gunflint iron covariation

Within the Gunflint biota, the preservation of internal cellular structure is coupled to early mineralisation. Morphospecies retaining cellular contents carry between 3×10^8 and 10^9 intracellular Fe atoms μm^{-3}, with intra-microfossil quartz finer than the surrounding matrix; empty sheaths remain at the chert background ($\sim 1.5 \times 10^6$ atoms μm^{-3}; Fig. 3) (Lepot et al., 2017a).

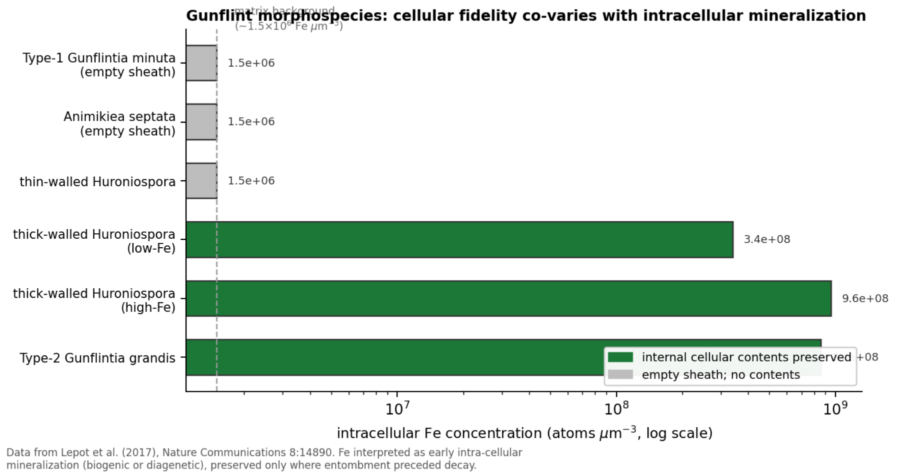

Figure 3: Intracellular Fe concentration in Gunflint morphospecies, against the preservation of cellular contents. The iron may reflect in vivo biomineralisation (Lepot et al., 2017a), so the covariation links fidelity to early mineralisation without isolating silica as the agent. Data from Lepot et al. (2017a).

This correlation carries a caveat that bears directly on the argument. Lepot et al. (2017a) interpret the intracellular iron as consistent with in vivo biomineralisation, plausibly by oxygenic photosynthesisers, rather than as a purely external precipitate. If so, the mineral phase associated with fidelity is partly a metabolic product, and the opposition between preservation and biology is less clean than a taphonomic reading would prefer. The observation nevertheless establishes the principle that fidelity required mineralisation before decay. It does not isolate silica as the agent; the assignment of silicification as the entombment medium rests on silica-specific evidence: the three-dimensional preservation of Gunflint cells within quartz, requiring silica encapsulation before compaction (Lepot et al., 2017a; Alleon et al., 2016a), and wall-conforming microquartz more generally (Wacey et al., 2012).

A second, structural confound limits what the within-Gunflint correlation can establish. The morphospecies that retain cellular contents are thick-walled forms, and wall recalcitrance and the capacity to nucleate intracellular minerals are co-varying properties of the morphotype rather than independent consequences of entombment timing. The within-Gunflint covariation is therefore suggestive but cannot, by itself, distinguish entombment timing from morphotype recalcitrance as the cause. It is consequently treated here as supporting micro-scale evidence; the less confounded comparison is the cross-formation meta-analysis of Section 4, where silicification varies at the facies level and is independent of morphotype.

5.1 Why redox and the silica cycle do not supply a clean GOE control

Iron speciation indicates pervasive ferruginous deposition on both sides of the GOE (Canfield, 1998; Poulton & Canfield, 2011), and the sulfur isotope signal of rising sulfate extends into the Neoarchean (Reinhard et al., 2009; Philippot et al., 2018). The sulfate rise bears on microbial sulfate reduction and the sulphurisation of organic matter, affecting the survival of bulk kerogen rather than the templating of cellular morphology. The capacity for early silica precipitation was likewise unchanging across the GOE (Siever, 1992; Perry & Lefticariu, 2007). The residual GOE signal identified by the partial correlation is therefore not readily attributed to any single measured redox or silica variable, and its origin remains open.

6 A case that cannot adjudicate: pre-GOE silicified cherts

The most promising candidate for a sharp test of the silicification hypothesis would be a pre-GOE chert that is demonstrably early-silicified. The ~3.46 Ga Apex chert and the ~3.48 Ga Dresser Formation are pre-GOE, silicified, and rich in cell-like carbonaceous structures, and so might appear to provide exactly such a test. They cannot. The Apex filaments described as microfossils (Schopf, 1993) were reinterpreted as abiogenic mineral artefacts (Brasier et al., 2002), and high-resolution reinvestigation has reinforced an abiogenic, hydrothermal origin for many such structures (Wacey et al., 2014). Where biogenicity is itself contested, no observation of cellular fidelity can adjudicate the entombment-timing hypothesis, because the question of whether these structures were ever cells is logically upstream of the question of how faithfully their morphology was preserved. A chert of disputed biogenicity is therefore structurally incapable of testing the hypothesis, and we treat Apex

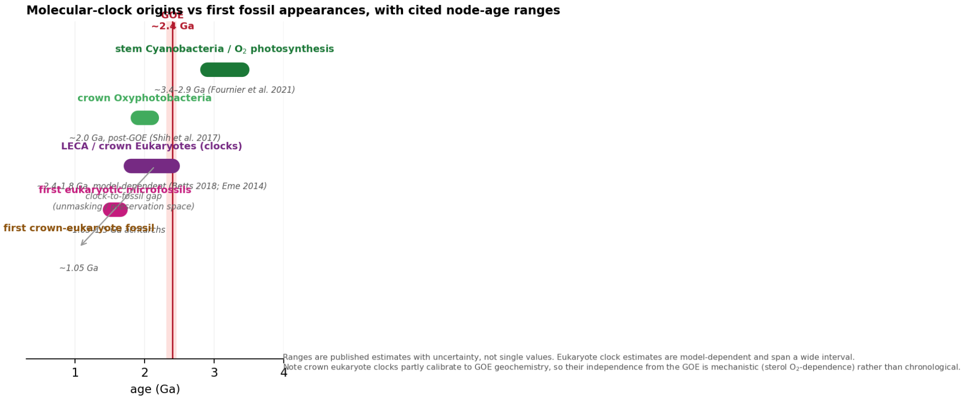

Figure 4: Molecular clock estimates of lineage origin against first fossil appearances, with published node-age ranges. The stem origin of cyanobacteria lies in the Archaean (Fournier et al., 2021); crown Oxyphotobacteria diversify around or after the GOE (Shih et al., 2017); eukaryote estimates are model-dependent (Betts et al., 2018; Eme et al., 2014) and partly calibrated to GOE geochemistry.

and Dresser as informative about the nature of silicified cherts rather than as a falsification test that the hypothesis has survived.

The observation these cherts do support is narrower but worth stating plainly. Silicification preserves morphology indiscriminately — whether the substrate is biological or abiotic — so silicified cherts will contain cell-like structures of mixed origin, and the biogenicity question is upstream of the fidelity question. The scarcity of unambiguous pre-GOE microfossils is consequently not evidence that silicification was feeble in the Archaean (it demonstrably was not (Siever, 1992; Perry & Lefticariu, 2007)), but that silicification records both biology and abiotic look-alikes, leaving few structures that meet modern biogenicity criteria.

7 Origins, first appearances, and the sterol constraint

Molecular clocks place the stem origin of cyanobacteria, and oxygenic photosynthesis, in the Archaean (Fournier et al., 2021), whereas the crown-group diversification of extant Oxyphotobacteria postdates the rise of oxygen (Shih et al., 2017). Molecular clock estimates for the last eukaryotic common ancestor are model-dependent but several precede the first widely accepted eukaryotic microfossils (Betts et al., 2018; Eme et al., 2014) (Fig. 4). The biomarker record cannot resolve the resulting gap: the steranes and 2-methylhopanes reported from ∼2.7 Ga shales (Brocks et al., 1999) reside in younger carbonate veins (Rasmussen et al., 2008) and are indistinguishable from procedural blanks in ultraclean cores (French et al., 2015).

The obligate aerobicity of sterol synthesis (Gold et al., 2017; Desmond & Gribaldo, 2009) cuts across a purely preservational reading for eukaryotes: crown eukaryotes have a mechanistic reason to diversify once oxygen became available, regardless of preservation. The clock-to-fossil gap for eukaryotes is thus not necessarily a preservational lag and may reflect genuine, oxygen-limited diversification. This supplies one candidate biological interpretation of the residual GOE signal identified in Section 4: if eukaryote and cyanobacterial diversification were oxygen-limited, the post-GOE rise in fidelity could partly record real biological change rather than preservation alone. The unmasking hypothesis is consequently better supported for cyanobacteria, whose metabolism clocks to the Archaean, than for sterol-synthesising eukaryotes, and any reading must be lineage-conditional.

8 Discussion

The meta-analysis is consistent with the hypothesis that silicification is a dominant predictor of cellular fidelity across the Precambrian microfossil record ($\rho = 0.87$, $p < 0.001$), but a GOE-associated residual signal in fidelity persists after controlling for silicification (partial $r = 0.72$; bootstrap 95% CI: 0.13–0.94). The defensible position is therefore narrower than a claim that silicification alone explains the post-GOE pattern. The preservation of cellular morphology in Precambrian cherts is conditioned by early mineralisation, predominantly silicification, that operated throughout but was expressed unevenly, depending on depositional facies and the timing of cementation relative to decay (Fig. 5). Where entombment was early and complete, morphology survives; where it was weak, kerogen and mats remain. Oxygenation plausibly modified organic-matter decay and kerogen sulphurisation (Lepot et al., 2009; Decraene et al., 2021) but did not gate the silicification on which cellular

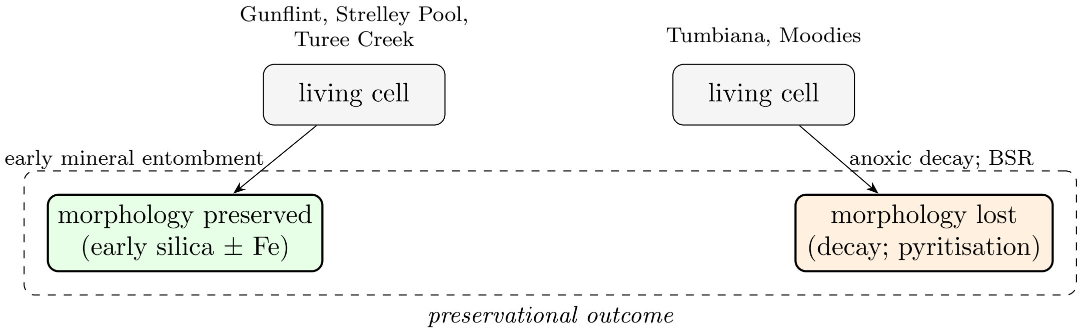

Figure 5: Preservational pathways. Where early mineral entombment outpaces decay, morphology is retained; where decay precedes mineralisation, morphology is lost and kerogen survives.

fidelity primarily depends. The residual GOE signal, however, is not absorbed by silicification, and its origin — biological diversification, an unmeasured preservational change, or sampling bias — cannot be resolved from this compilation alone.

The limitations of the compilation must be stated plainly, since they define the status of the contribution. The silicification and fidelity scores are ordinal and were assigned from the descriptive literature; they are not source-independent, because the petrographic and fidelity assessments alike are drawn from a literature dominated by a few overlapping research groups (principally Lepot, Javaux and Wacey). The partial-correlation residual is sensitive to the scoring scheme and to the inclusion of the two GOE-spanning units. The within-Gunflint iron covariation is single-locality, single-element, possibly biogenic, and confounded by wall recalcitrance. None of these limitations undermines the principal observation — that the silicification–fidelity association is robust in direction — but they do bound the strength of the claim about the residual GOE signal, whose biological versus preservational origin the meta-analysis cannot decide.

One potential confound that the compilation controls for by design is thermal maturity. All 13 formations in the compilation are of low metamorphic grade, ranging from sub-greenschist to prehnite–pumpellyite facies. Thermal maturity therefore has effectively no variance across the sample and cannot account for the observed range in cellular fidelity (1–3 within the low-grade subset). This control-by-design is a strength of the compilation, though it also means the analysis cannot assess whether the hypothesis holds among higher-grade successions.

The hypothesis advanced here would be falsified by (i) a pre-GOE chert bearing confirmed genuine microfossils with demonstrably early silicification and poor cellular fidelity, or (ii) a within-formation regression showing no relationship between fidelity and a quantitative silicification index scored blind and across research groups. Neither is known; both are testable, and the second is achievable with existing material. The programme needed to pursue them is specified below.

The implications are qualified accordingly. Abundance and diversity counts across the GOE cannot be read as proxies for biomass without facies correction. The clock-to-fossil gap is consistent with a preservational lag but, for sterol-synthesising lineages, may equally reflect oxygen-limited diversification — a reading that is now lent indirect support by the persistence of a GOE signal after silicification is controlled for. For the search for ancient biosignatures, the corollary is that the preservation of morphology requires a silica-precipitating facies rather than the presence of oxygen, but the appearance of morphological diversity after the GOE should not be attributed to preservation alone.

8.1 What would be needed: an experimental and analytical programme

The hypothesis is testable; we specify three approaches in order of decisiveness.

(a) Controlled silicification–decay experiments. The confound that most limits the within-Gunflint evidence is the co-variation of wall recalcitrance and mineral-nucleation potential with morphotype. A laboratory programme can hold morphotype constant and vary only the timing of silica addition. Microbial cultures spanning thick- and thin-walled morphotypes (e.g. cyanobacteria and other bacteria) would be incubated under factorial O_2 × sulfate × Fe conditions, with controlled silica added at defined intervals after death. Blind-scored cellular fidelity, silica nucleation timing, and intracellular mineralisation would then reveal whether entombment timing — rather than wall recalcitrance — is the causal variable. The experimental demonstration that early silica entombment minimises molecular degradation during advanced diagenesis (Alleon et al., 2016b) provides the mechanistic foundation for such a programme; what is needed is the controlled timing manipulation that no published experiment has yet performed.

(b) A blind, cross-group within-formation regression. This is the test the present compilation proposes but does not perform at the population level. Cellular fidelity and a quantitative silicification index would be scored independently by different research groups, blind to one another's scores, on 30–50 microfossil populations per formation, and regressed while controlling for thermal maturity (Raman R_2). A null relationship across formations of comparable grade would weigh against the hypothesis; a consistent positive relationship would support it.

(c) A systematic survey for the falsifier. A single pre-GOE chert (or indeed any-age chert) carrying microfossils of undisputed biogenicity, demonstrably early silicification, and poor cellular fidelity would falsify the hypothesis outright. A targeted search of curated chert collections, screened first for biogenicity and silicification timing and then scored blind for fidelity, is the most economical route to a decisive test. No such case is currently known.

9 Conclusions

We set out to advance and evaluate the hypothesis that early mineral entombment, predominantly silicification, is a dominant taphonomic confound in the apparent post-GOE diversification of microfossils. The hypothesis is supported, but not confirmed. The strongest evidence is kinetic: using the Rimstidt & Barnes (1980) rate law, abiotic silica precipitates a 100 nm entombment layer in $\approx$1.3 hours under Precambrian silica saturation ($\Omega \approx 1.8$), one to two orders of magnitude faster than anoxic decay (3–14 days), establishing feasibility with quantified uncertainty from surface catalysis and inhibition. The meta-analysis is concordant in direction: across 13 biotas of accepted biogenicity, silicification predicts cellular fidelity (Spearman $\rho = 0.87$, $p < 0.001$), and this association is evaluated with a permutation test ($p = 0.0006$), a null-model comparison ($p = 0.0002$), a $\pm$1-point perturbation analysis (positive in 95.2% of 10,000 trials), and an off-diagonal discrimination test (3 of 13 formations plot away from the silicification–fidelity diagonal). The observed $\rho = 0.87$ should, however, be treated as an upper bound, inflated by clean coding, rather than a point estimate. GOE position retains a residual association with fidelity after controlling for silicification (partial $r = 0.72$; bootstrap 95% CI: 0.13–0.94), positive in 98.5% of bootstraps — robust in direction but very uncertain in magnitude. Thermal maturity is controlled by design: all 13 formations are of low metamorphic grade and cannot vary enough to explain the fidelity range. Whether the GOE residual reflects genuine biological diversification, an unmeasured preservational change, or sampling bias remains unresolved by the meta-analysis. The biomarker evidence for pre-GOE eukaryotes is contaminated, but the obligate aerobicity of sterol biosynthesis gives eukaryotes a mechanistically independent reason to diversify with oxygen — one candidate biological interpretation of the residual. We state the two conditions that would falsify the silicification hypothesis and specify the experimental and analytical programme needed to test them, one element of which is achievable on existing material. Confirmation awaits that programme; the present paper makes the case that the hypothesis is worth testing.

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